

2.9 Dividing Polynomials

Lesson Introduction

 Benchmark	MA.912.AR.1.4 Divide a polynomial expression by a monomial expression with rational number coefficients.		 Learning Target	I can divide a polynomial expression by a monomial expression.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.AR.1.3 Rewrite the sum of two algebraic expressions that have a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.		MA.912.AR.1.6 Solve mathematical and real-world problems involving addition, subtraction, multiplication, or division of polynomials. MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real or complex number system.	
 Essential Questions	- How can I divide a polynomial expression by a monomial expression? - Is the quotient of a polynomial and a monomial always a polynomial?			
 Lesson Narrative	Students will conclude this unit by learning how to divide polynomials by monomials. Students will begin by dividing monomials by monomials and then make connections to dividing polynomials by monomials using the area model. They will learn that the result of dividing a polynomial by a polynomial will not always result in a polynomial.			
 Component Summary	2.9.1 Warm-Up (~5 min.)	Bell work adding, subtracting, and multiplying polynomials (<i>SE p. 101</i>)	2.9.3 Guided Practice (10 min.)	Practice dividing polynomials by monomials (<i>SE p. 103</i>)
	2.9.2 Guided Instruction (10-15 min.)	Divide polynomials by monomials and establish that division of polynomials is not a closed operation (<i>SE p. 101</i>)	2.9.4 Your Turn! (5-10 min.)	Dividing polynomials by monomials (<i>SE p. 104</i>)
	2.9.5 Wrap-Up (~5 min.)	Reflecting on the lesson: what is understood, three points, and what questions still linger (<i>SE p. 104</i>)		
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Divisor - Dividend - Quotient	N/A	N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may forget to apply the appropriate law of exponents when dividing. - Students may not divide each term in the polynomial when dividing. - Students may make errors with the signs of the numbers when dividing.	The students have not explored division of polynomials yet. They did engage with division of monomials using the laws of exponents earlier in the unit. Consider spending a few minutes reviewing and connecting that concept to the process they learn in this lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect In 2.9.2 Guided Instruction, students receive formal instruction on how to divide a polynomial by a monomial. Leverage the Compare and Connect routine as a method to demonstrate teacher thinking out loud. This can support meta-cognitive and meta-linguistic awareness.	MA.K12.MTR.3.1: Mathematical Fluency Students will work in this lesson to divide a polynomial by a monomial while maintaining flexibility and accuracy. The goal is for students to complete tasks accurately and with confidence. This includes applying laws of exponents with fluency.
 Differentiate Instruction	For this lesson, consider the use of an Anchor Chart to create a point of reference to assist students with dividing polynomials. Consider including a vocabulary component to highlight the divisor, dividend, and quotient from a polynomial division problem.	
	For students who benefit from the area model, consider preparing blank models that students can use throughout the 2.9.3 Guided Practice and 2.9.4 Your Turn!	
Lesson Notes:		

2.9.1 Warm-Up: Bell Work

Component Overview: Students will complete a Bell Work on adding, subtracting, and multiplying polynomials.



2.9.1 Warm-Up: Bell Work

Simplify each expression.

1. $1.3u(4u^5 + 7t)$
 $5.2u^6 + 9.1tu$

2. $\left(\frac{7}{10}g^3 + \frac{12}{5}g^2\right) + (35g^4 + 6g^2 - 10g)$
 $35g^4 + \frac{7}{10}g^3 + \frac{42}{5}g^2 - 10g$

3. $\left(7x^3y - \frac{49}{10}xy\right) - \left(-\frac{11}{10}x^3y + \frac{3}{5}xy + 7xy^2\right)$
 $\frac{81}{10}x^3y - \frac{11}{2}xy - 7xy^2$



Use the data collected from this activity to determine potential remediation steps prior to wrapping up this unit.

2.9.2 Guided Instruction: Dividing Polynomials by Monomials

Component Overview: Students will learn how to divide polynomials by monomials and determine whether dividing a polynomial by a monomial results in a quotient that is a polynomial.



2.9.2 Guided Instruction: Dividing Polynomials by Monomials

Multiplication and division are inverse operations. A polynomial can be multiplied by a monomial. Therefore, it can also be divided by one.

1. Consider the expression $7m(m + 7)$.
 - a. Determine the product of $7m(m + 7)$.
 $7m^2 + 49m$
 - b. Write two division equations using the factors $7m$ and $(m + 7)$, and the product of $7m(m + 7)$ found in part a.

$$7m^2 + 49m \div 7m = (m + 7)$$

$$7m^2 + 49m \div (m + 7) = 7m$$

2. Label each part of the mathematical statement $(12x^4 - 16x^3) \div 4x^2 = 3x^2 - 4x$ using words from the bank.

Word Bank

divisor

dividend

quotient

<i>divisor</i>	<i>quotient</i>	<i>dividend</i>
$4x^2$	$3x^2 - 4x$	$12x^4 - 16x^3$



Draw attention to the inverse operation vocabulary used here to connect multiplying polynomials to the division of polynomials. This can help with the conceptual understanding of how the two operations connect. Students can also then use their understanding of multiplying polynomials to assist as they learn to divide.



For English Language Learners, consider adding additional translations of the words on an Anchor Chart with the others so they can connect to the operational vocabulary in a way that makes it easier to understand.

2.9.2 Guided Instruction: Dividing Polynomials by Monomials (continued)

3. Consider the polynomial $6p^4 - 8p^3 + 2p^2$.

- a. Determine the quotient of each term of the polynomial $6p^4 - 8p^3 + 2p^2$ and $2p^2$ where $p \neq 0$.

$$6p^4 \div 2p^2 = 3p^2 \quad -8p^3 \div 2p^2 = -4p \quad 2p^2 \div 2p^2 = 1$$

- b. Use the quotients found in part a to determine the quotient of $(6p^4 - 8p^3 + 2p^2) \div 2p^2$, where $p \neq 0$ using the area model.

	$3p^2$	$-4p$	1
$2p^2$	$6p^4$	$-8p^3$	$2p^2$

$$(6p^4 - 8p^3 + 2p^2) \div 2p^2 = 3p^2 - 4p + 1$$

- c. Discuss with your partner if the quotient is a polynomial.

4. Division of a polynomial can be represented as a fraction.

- a. The division of a polynomial by a monomial is shown, where $p \neq 0$. Simplify each term by using the quotient of powers law.

$$\frac{(6p^4 - 8p^3 + 2p^2)}{2p^2} = \frac{6p^4}{2p^2} - \frac{8p^3}{2p^2} + \frac{2p^2}{2p^2} = 3p^2 - 4p + 1$$

- b. Is the quotient a polynomial? Explain why or why not.

Yes, the quotient is a polynomial because each term is a constant, variable, or a product of a constant and variable, and each variable is raised to a nonnegative integral power.

- c. Why was it stated that $p \neq 0$?

When $p = 0$, it creates a situation where you are dividing by 0 which makes it undefined.



After reading the prompt for question 3a and before completing the table, discuss which of these polynomials is the dividend and which is the divisor.

For question 3b ask students: "What do you multiply $2p^2$ by to obtain $6p^4$? $-8p^3$? $2p^2$?"



Have students share their answer to whether they think it is a polynomial or not. Engage students in a polling protocol where they vote on whether they think it is or it is not, selecting students to provide rationale.

Note that students do not need to write a response here. Here they can discuss and will have an opportunity to write their thoughts on question 4.



Engage in discussion of why each term in the numerator (the dividend) should be divided by the term in the denominator (the divisor). Make connections to multiplying using the distributive property.

2.9.2 Guided Instruction: Dividing Polynomials by Monomials (continued)

5. Consider the expression $\frac{18b^4d + 12b^3d^2 - 24bd^3}{0.6b^2d}$.

a. Fill in the blanks in each step to determine the quotient, where $b \neq 0$ and $d \neq 0$.

$$\frac{18b^4d + 12b^3d^2 - 24bd^3}{0.6b^2d} = \frac{18b^4d}{0.6b^2d} + \frac{12b^3d^2}{0.6b^2d} - \frac{24bd^3}{0.6b^2d}$$

$$\frac{18b^4d}{0.6b^2d} + \frac{12b^3d^2}{0.6b^2d} - \frac{24bd^3}{0.6b^2d}$$



$$30b^2$$



$$+ 20bd$$



$$- \frac{40d^2}{b}$$

b. Discuss with your partner why the quotient is not a polynomial. Summarize your discussion.

Answers will vary. This is not a polynomial because the third term results in a variable in the denominator.

6. Complete that statement.

The result of

- ☐ adding
- ☐ subtracting
- ☐ multiplying
- ☒ dividing

a polynomial

- ☒ by
- ☐ from
- ☐ to

a polynomial will not always result in a polynomial.



Spend some time reviewing the answers to question 6. In Lesson 8, students learned about how adding, subtracting, and multiplying always results in a polynomial. Now they are learning that with division, that is not true.

2.9.3 Guided Practice: Dividing Polynomials by Monomials

Component Overview: Students will complete three practice questions with dividing polynomials by monomials.



2.9.3 Guided Practice: Dividing Polynomials by Monomials

1. Use the area model to find $(3ac + 2.1a^3c - 1.5ac^2) \div 3ac$, where $a \neq 0$ and $c \neq 0$.

	1	$0.7a^2$	$-0.5c$
$3ac$	$3ac$	$2.1a^3c$	$-1.5ac^2$

$$1 + 0.7a^2 - 0.5c$$

2. Find the quotient if $x \neq 0$ and $y \neq 0$.

$$\frac{56x^2y^2 - 16x^2y^4 - 48x^4y^3}{-8x^2y^4}$$

$$-\frac{7}{y^2} + 2 + \frac{6x^2}{y}$$

3. Divide $\left(\frac{4}{5}k^2 - \frac{2}{3}k\right) \div \left(-\frac{5}{6}k\right)$, where $k \neq 0$.

$$-\frac{24}{25}k + \frac{4}{5}$$



Watch for how students are approaching each of these questions. Students may forget to apply exponent laws when dividing or might make computational errors. Some students will attempt to do problems like question 2 in their head and make errors in that process or fail to divide each term in the numerator by the denominator.



Have students create a question for a partner to solve if time permits.

2.9.4 Your Turn!

Component Overview: Students will complete two practice questions independently with dividing a polynomial by a monomial. Use the data collected while monitoring student work to determine steps for remediation.



2.9.4 Your Turn!

Find each quotient.

1. $\frac{7h^7 - 91h^8}{7h^7}$, where $h \neq 0$.

$$1 - 13h$$



Consider the use of Think-Pair-Share for this activity. Allow the students to complete both questions on their own and then compare their work with a partner to check for accuracy and come to consensus on the correct answer, checking with other partners if needed.

2. $\left(\frac{1}{24}f^3r^2 - \frac{1}{4}f^4r - \frac{1}{2}f^2r\right) \div (-4f^2r)$, where $f \neq 0$ and $r \neq 0$

$$-\frac{fr}{96} + \frac{f^2}{16} + \frac{1}{8}$$



For students who were struggling in 2.9.3 Guided Practice, consider pulling a small group of students to work on question 1 together. This can help reinforce the content so they can complete question 2 on their own.

2.9.5 Wrap-Up: Reflection

Component Overview: Students will complete a reflection where they share what is now squared away, three points that are important, and something still circling in their brain as they think about the content from the entire unit.

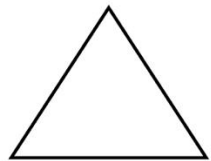


2.9.5 Wrap-Up: Reflection

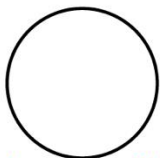
1. What is something that is now “squared away” in your mind about adding, subtracting, multiplying, and dividing polynomials?



2. What are three “points” that are important about adding, subtracting, multiplying, and dividing polynomials?



3. What is something that is still “circling” your brain about adding, subtracting, multiplying, and dividing polynomials?



Answers will vary.



The information provided from the student here, coupled with what was observed during instruction, can provide information for potential next steps for remediation as part of the unit review prior to assessment.



Consider pairing students to discuss the concepts still circling their brains to see if peers can answer any of their questions.